Supplementary Material: Winning Is Not Killing

Anonymous Author(s)

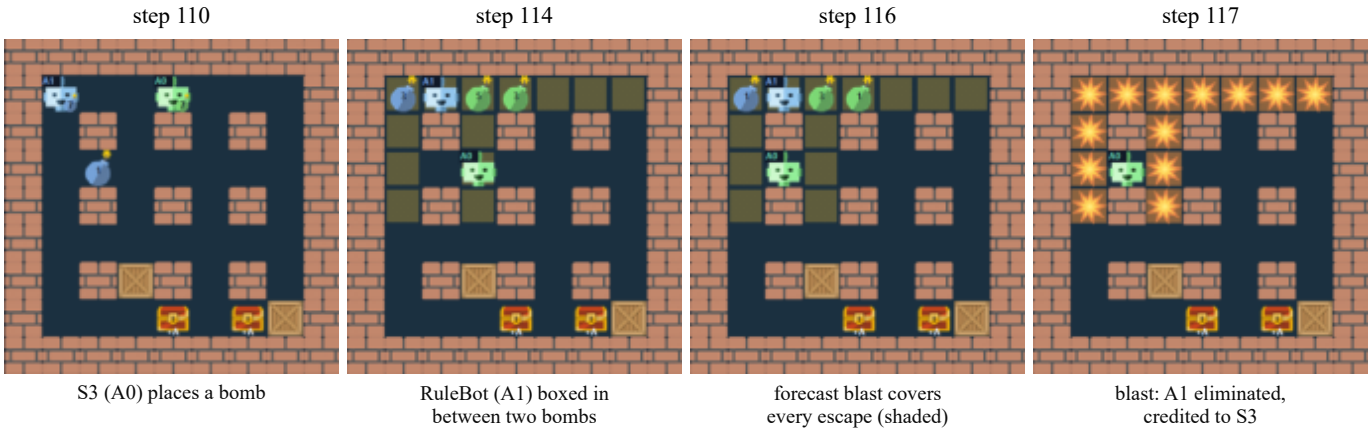


Fig. S1. Four frames of a real game: S3 (A0, green) against RuleBot (A1, blue) on validation board 21000. The number on a bomb is its fuse; shaded cells are the forecast blast area. S3's bombs at steps 110 and 112 box RuleBot in; the chain explosion at step 117 kills it and is credited to S3, so the game counts as an active win.

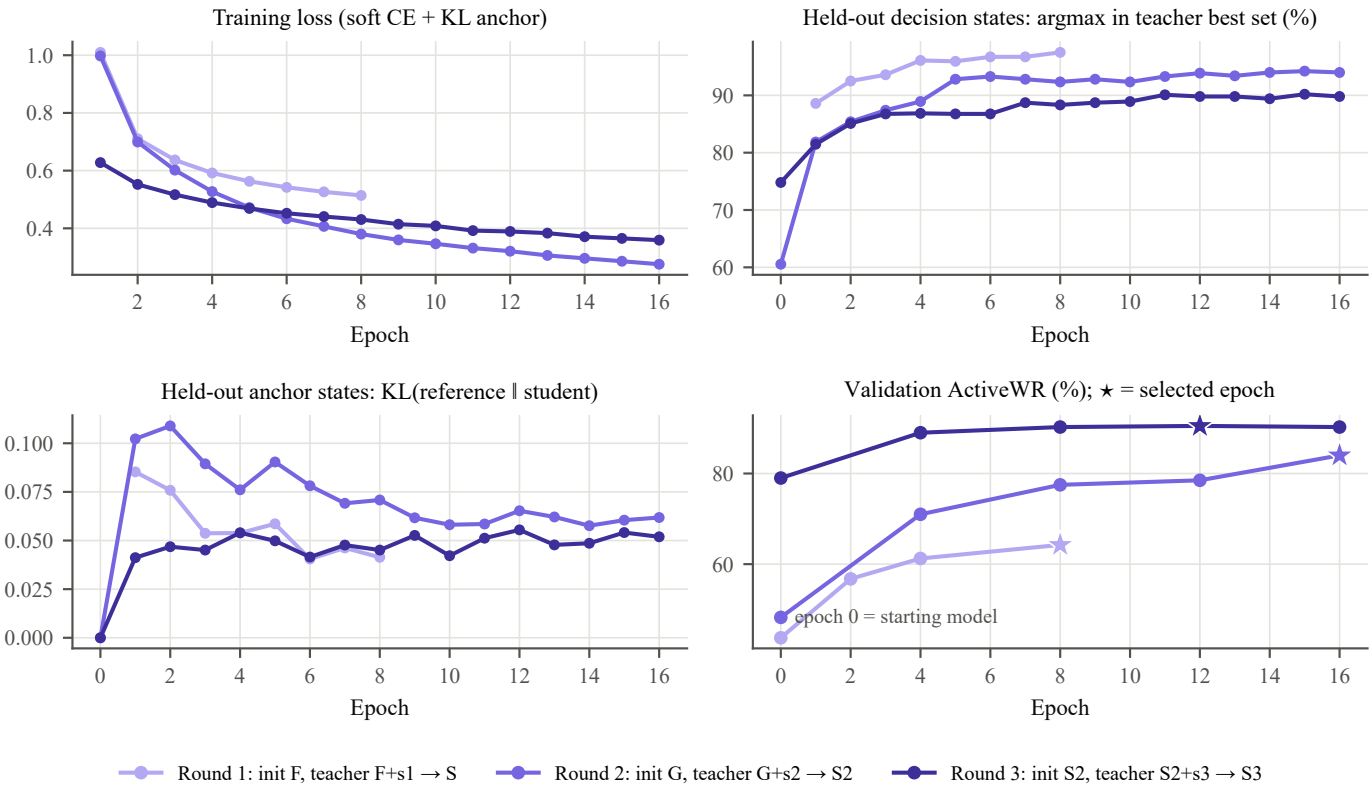


Fig. S2. Distillation rounds. Top: training loss and share of held-out decision states where the student's argmax is in the teacher's best set. Bottom: KL from the starting policy on held-out anchors, and validation ActiveWR (epoch 0 = starting model; stars = epochs selected before testing). Held-out sets differ between rounds.

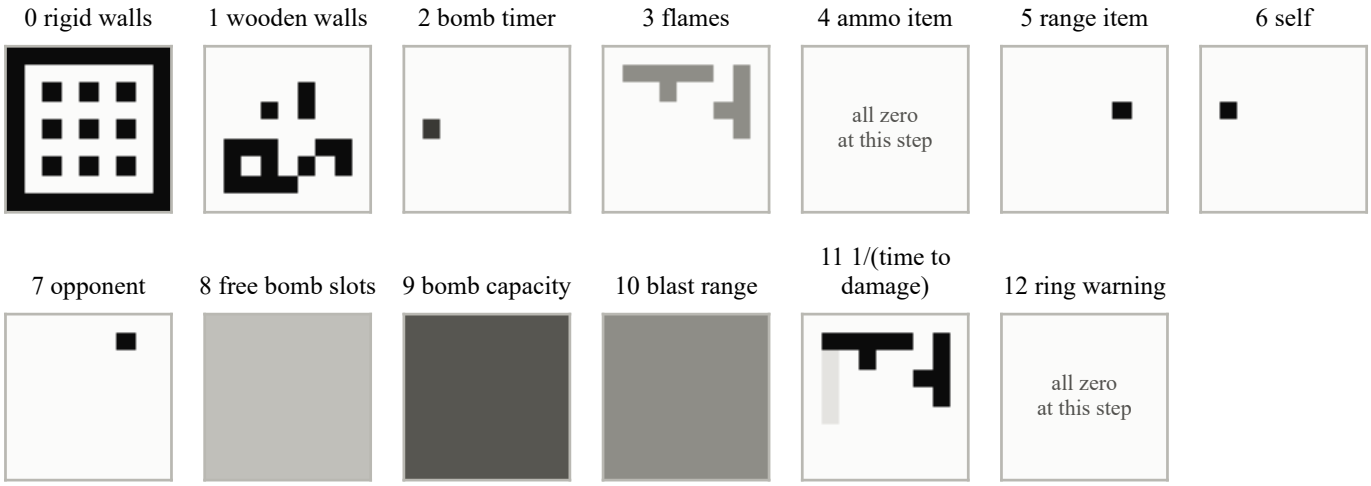


Fig. S3. The 13 observation planes for one agent (validation board 21007, step 26, RuleBot vs RuleBot; darker = larger). The agent has collected one ammo item (capacity 2 of 3) and has one bomb on the board, so one slot is free; a range item lies on the board. Planes marked "all zero" are empty at this step: no ammo item is visible, and the ring warning starts 40 steps before the first closure.

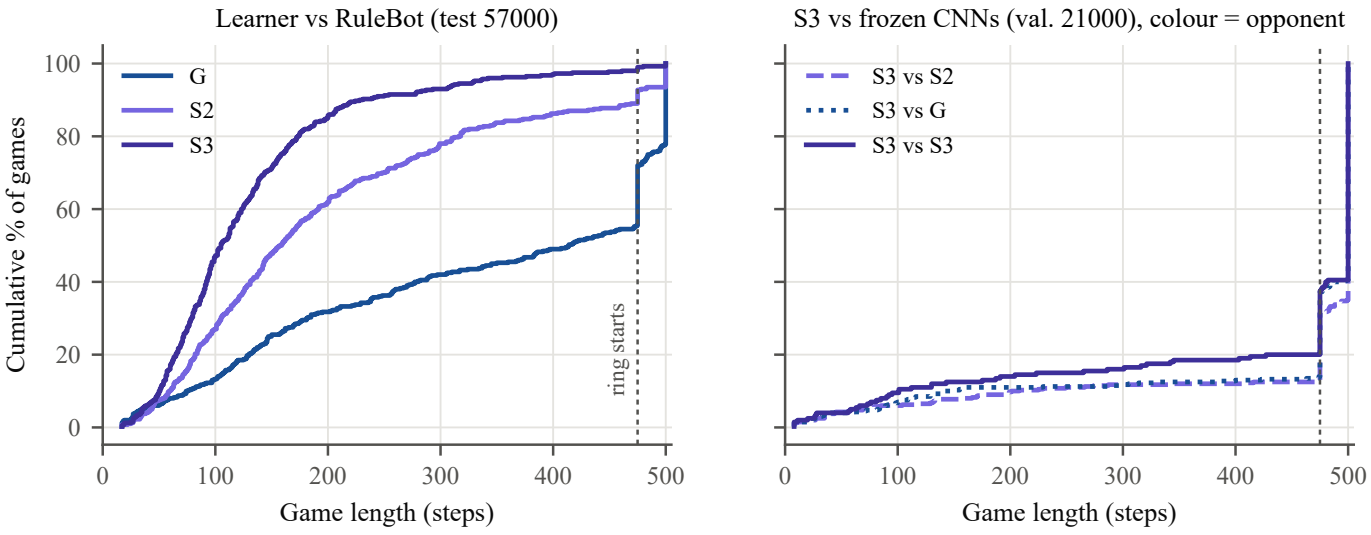


Fig. S4. Cumulative distribution of game length. Against RuleBot, stronger agents finish earlier (median 414, 158 and 105 steps for G, S2, S3; 22.25, 6.5 and 0.75% reach the limit). Against frozen CNNs (right; colour = opponent), 59.5–65% of S3's games reach 500 steps. Left: test block 57000; right: validation block 21000.

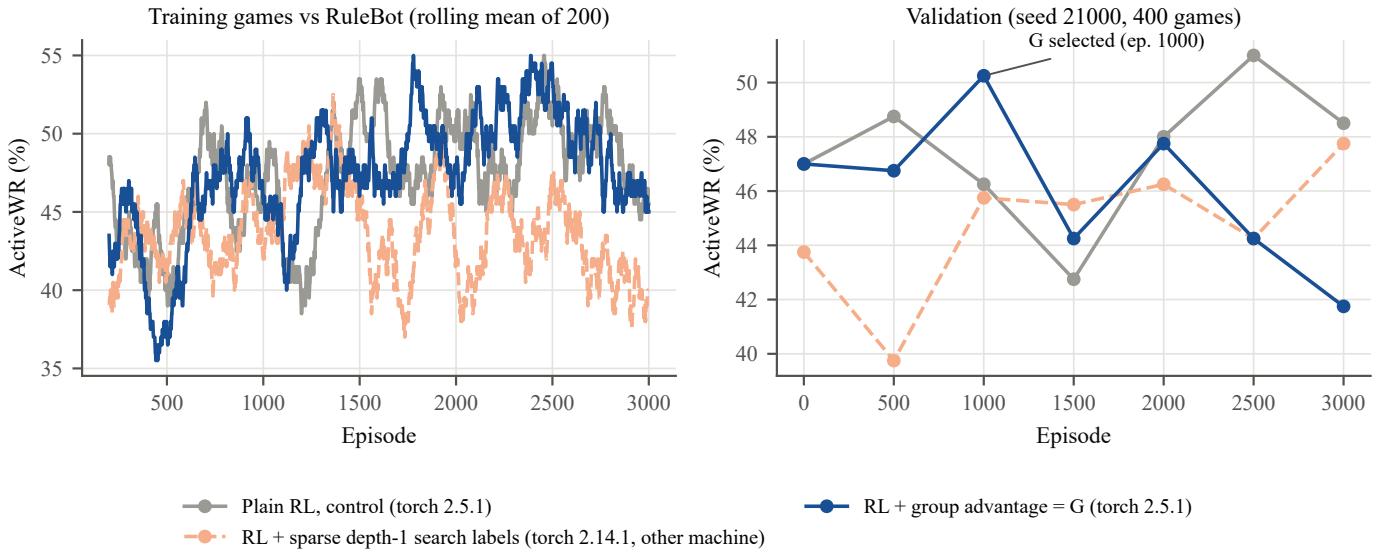


Fig. S5. RL from F (3,000 episodes per arm, one train seed). Left: rolling active win rate of training games against RuleBot. Right: validation active win rate every 500 episodes. The group-advantage arm peaked at episode 1000 (50.25) and became G; on the test block it was +0.75 over the control. The dashed arm ran on a second machine with another torch version, so its validation numbers are not comparable with the other two.